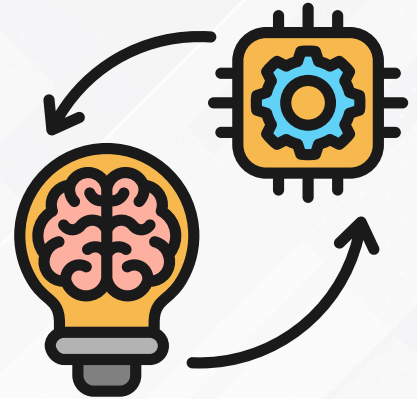


MASTER AI FORWARD DEPLOYMENT ENGINEER

End-to-End Solution Deployment





SKILLS YOU WILL LEARN



Programming & Scripting

Python, Bash, REST APIs



Data Engineering

SQL, ETL/ELT, Data Pipelines



Cloud Platforms

AWS, Azure, GCP



Containerization

Docker, Kubernetes



CI/CD & Automation

Jenkins, GitHub Actions



Model Deployment

FastAPI, Flask, TensorFlow Serving



Monitoring & Logging

Prometheus, Grafana, ELK Stack



Security & Governance

IAM, Secrets, Compliance, Best Practices



APPLICATIONS / WHERE YOU WILL WORK



Banking & Financial Services

Deploy credit risk models, fraud detection, loan underwriting, collections optimization



E-commerce & Retail

Recommendation engines, demand forecasting, customer segmentation, dynamic pricing



Telecom

Churn prediction, network optimization, usage analytics, fraud detection



Healthcare

Patient risk scoring, readmission prediction, operational efficiency, image AI deployment



Manufacturing

Predictive maintenance, quality inspection, production optimization



Logistics & Supply Chain

Route optimization, inventory forecasting, delay prediction



Government & Public Sector

Citizen services, policy impact modeling, analytics-driven decision making

Student Benefits:



GITHUB PROFILE
CREATION &
PROFILE BUILDING



RESUME
BUILDING



TESTS & MOCK
INTERVIEWS



LIVE
CLASSES



LMS
ACCESS



HANDS-ON
PROJECT



INTERVIEW
PREPARATION



INTERVIEW
PREPARATION



INDUSTRY
MENTORSHIP



PLACEMENT
SUPPORT



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What is a Forward Deployed Engineer (FDE)?

What is a Forward Deployed Engineer (FDE)?

A **Forward Deployed Engineer (FDE)** is a hybrid **engineer + consultant** who works directly with a client to build and ship a working solution inside the client's real environment — not a generic product in a lab.

An FDE sits at the intersection of three worlds:

- **Engineering** – they write real code, integrate systems, deploy to production, and debug live.
- **Consulting** – they talk to stakeholders, translate messy business problems into technical requirements, and manage expectations.
- **AI** – they know how to apply modern AI (LLMs, RAG, agents) to solve those problems fast.

The core difference: A regular Software Engineer builds a product for many users. An FDE builds a solution for one client, on-site (or embedded), tuned to that client's data, constraints, and goals.

What makes a great FDE:

- Moves fast – ships a working demo in hours/days, not months.
- Prioritizes **speed over perfection and empathy over assumptions**.
- Is full-stack across **AI + Data + Cloud + Software Engineering** (the four pillars).
- Can talk to a CTO and a data analyst in the same afternoon and be understood by both.



MODULE 1

What is a Forward Deployed Engineer? (5 lessons)



- 1.1 What is an FDE? | The intersection of engineering, consulting, and AI
- 1.2 FDE vs SWE | FDEs build for clients, SWEs build products
- 1.3 The FDE Mindset | Speed > Perfection · Empathy > Assumptions
- 1.4 FDE Skillset | AI + Data + Cloud + SE = the four pillars
- 1.5 Delivery Method | Concepts, MCQs, and Labs

MODULE 2

AI & GenAI Foundations (4 lessons)



- 2.1 Generative AI & the AI → ML → DL → GenAI evolution | Models that generate content, not just classify
- 2.2 Transformers, Tokens & Tokenization | Simplified mental model; why tokens drive cost & limits
- 2.3 Customizing AI: Pre-training vs Fine-tuning vs RAG · Open vs Closed models | When to use each
- 2.4 AI Application Lifecycle | Idea → Prototype → Eval → Production → Monitor

MODULE 3

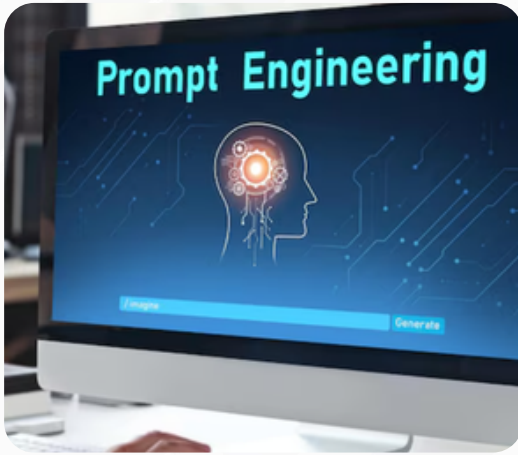
Working with LLMs (4 lessons)



- 3.1 How LLMs Work & Context Windows | Next-token prediction; the LLM's "RAM" — what fits, what gets cut
- 3.2 Hallucination | Why LLMs make things up; mitigation strategies
- 3.3 Model Selection for FDEs | Claude / GPT-4o / Gemini / LLaMA; the cost-speed-quality triangle
- 3.4 API vs Self-Hosted, Cost & Latency | Token pricing, batching, caching; compliance & scale

MODULE 4

Prompt Engineering Essentials (4 lessons)



- 4.1 Prompt Engineering Overview | 80% of AI quality comes from prompts
- 4.2 Core Techniques | Zero-shot, Few-shot, Chain-of-Thought & Prompt Chaining
- 4.3 System Prompts, Structured Output & Controls | Persona + constraints; JSON mode; temperature
- 4.4 Evaluation & Enterprise Patterns | A/B testing; classification, extraction, summarization, transformation

MODULE 5

RAG & Vector Databases (5 lessons)



- 5.1 RAG Foundations | Retrieval-Augmented Generation, embeddings & vector databases
- 5.2 Building a RAG Pipeline | Chunking, similarity search, ingest → embed → store → retrieve → generate
- 5.3 Retrieval Strategies & Metadata Filtering | Top-K, MMR, re-ranking; filter before vector search
- 5.4 Advanced RAG | Self-query, HyDE, Corrective RAG, RAPTOR
- 5.5 RAG Evaluation | Faithfulness, Answer Relevance, Context Recall (RAGAS)

MODULE 6

AI Agents & Orchestration (5 lessons)



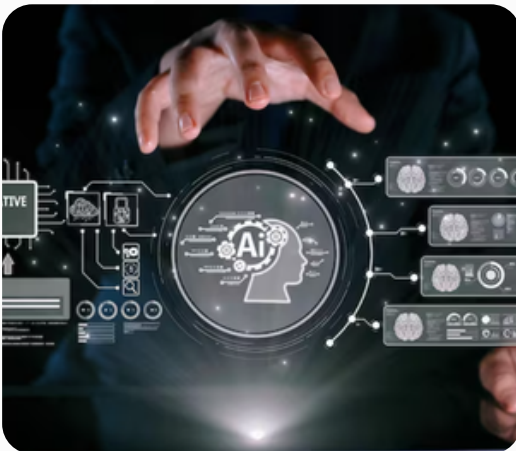
- 6.1 What is an Agent? | LLM + Tools + Memory + Decision Loop; tool calling via JSON schema
- 6.2 Agentic Patterns | ReAct (Reason → Act → Observe → Repeat), planning, reflection, memory
- 6.3 Multi-Agent Systems & Frameworks | Supervisor-worker, peer agents; LangChain & LangGraph
- 6.4 Agent Observability | Trace every step; log tool calls; measure token usage
- 6.5 Lab: Research Agent

Goal — build an agent that:

1. Accepts a research question
2. Searches the web · tool: web_search
3. Reads relevant pages · tool: read_url
4. Synthesizes findings into a structured report
5. Saves output to a file · tool: write_file

MODULE 7

MCP & AI Workflow Automation (4 lessons)



- 7.1 What is MCP? | Standard protocol for LLMs to call external tools safely; MCP vs REST API
- 7.2 MCP Components | Host (Claude Desktop), Client (app), Server (tool wrapper)
- 7.3 Building an MCP Server | Expose any tool (DB, API, file system) via MCP
- 7.4 AI Workflow Design & Automation | Directed graphs + human checkpoints; trigger, event-driven & scheduled patterns

MODULE 8

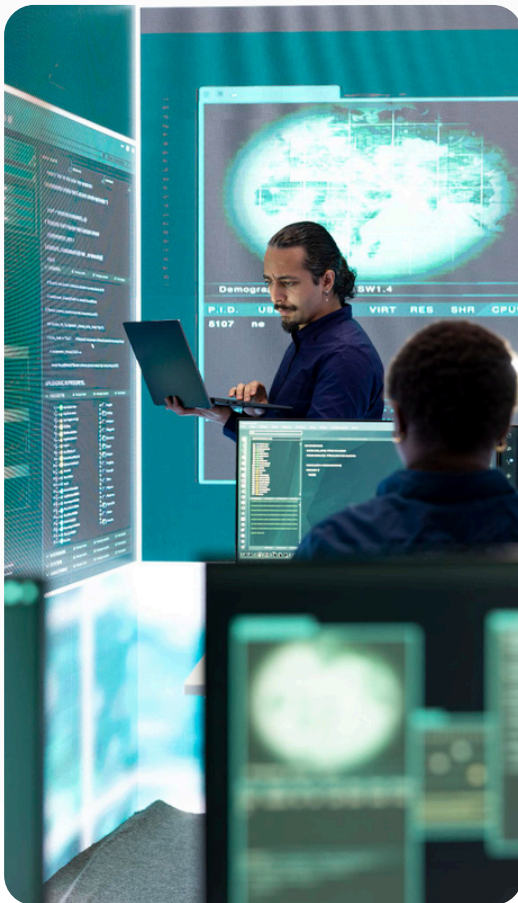
AI-Assisted Software Engineering (8 lessons)



- 8.1 AI-First SDLC | Prompt → Generate → Review → Refine → Ship (not: write → debug → repeat)
- 8.2 Claude Code | Agentic CLI that reads/writes files, runs commands, fixes bugs
- 8.3 Cursor | AI-native IDE — inline generation, codebase-aware completions
- 8.4 GitHub Copilot | Autocomplete + chat — best for line-by-line acceleration
- 8.5 Rapid Prototyping | 0 → working demo in 2 hours using AI tools
- 8.6 AI Code Review | Use AI to review your own PRs before submitting
- 8.7 AI Debugging | "Here's the error + stack trace — what's wrong?"
- 8.8 Lab: Build a Full Feature in 2 Hours Using AI Tools

MODULE 9

Data Engineering & System Design for AI (12 lessons)



- 9.1 DE Fundamentals | Data is the fuel for AI — FDEs must be able to wrangle it
- 9.2 ETL vs ELT | Modern: load first, transform in the warehouse (ELT wins)
- 9.3 Data Lake vs Lakehouse | Lakehouse = Delta Lake on object storage — best of both
- 9.4 Medallion Architecture | Bronze (raw) → Silver (cleaned) → Gold (business-ready)
- 9.5 Batch vs Streaming | Batch for history; streaming for real-time AI
- 9.6 Data Pipelines | Ingestion → transformation → loading → validation
- 9.7 Data Quality | Null checks, schema validation, freshness monitoring
- 9.8 Platform Landscape | Databricks, Snowflake, dbt, Airflow, Kafka — when to use each
- 9.9 AI System Design Fundamentals | Requirements → components → data flow → interfaces for an AI solution
- 9.10 Designing the AI Data Layer | Feature stores, vector stores, caches & how AI pipelines consume data
- 9.11 Scalability & Reliability Patterns | Load balancing, queues, retries, idempotency, graceful degradation
- 9.12 System Design Case Study | Architecting an end-to-end enterprise RAG/agent system

MODULE 10

Cloud Architecture & AI Deployment (14 lessons)



- 10.1 Cloud-Native Principles | Design for failure; scale horizontally; automate everything
- 10.2 AWS Well-Architected | 5 pillars every FDE should know before touching client cloud
- 10.3 AI Platform Deployment | Where AI apps live: EC2, Lambda, ECS, Bedrock, SageMaker
- 10.4 Microservices & APIs | Decompose AI apps into independently deployable services
- 10.5 Docker for AI | Container your AI app → consistent, portable, deployable
- 10.6 Kubernetes for AI Workloads | Orchestrate, scale & self-heal containerized AI services
- 10.7 Serverless for AI | Lambda for lightweight inference; cost-effective for event-driven AI
- 10.8 Model Serving & Inference Infrastructure | Endpoints, autoscaling, GPU vs CPU, batching for throughput
- 10.9 Deployment Strategies | Blue-green, canary & rolling releases for AI services
- 10.10 Infrastructure as Code | Terraform / CloudFormation — reproducible, version-controlled environments
- 10.11 CI/CD for FDEs | GitHub Actions → test → build → deploy on every merge
- 10.12 Observability & Monitoring in Production | Logs, metrics, tracing, latency, cost & drift dashboards
- 10.13 Security Fundamentals | IAM least-privilege; VPC isolation; secrets in AWS Secrets Manager
- 10.14 Lab: Deploy an AI Service End-to-End | Container → cloud → CI/CD → monitored production endpoint

MODULE 11

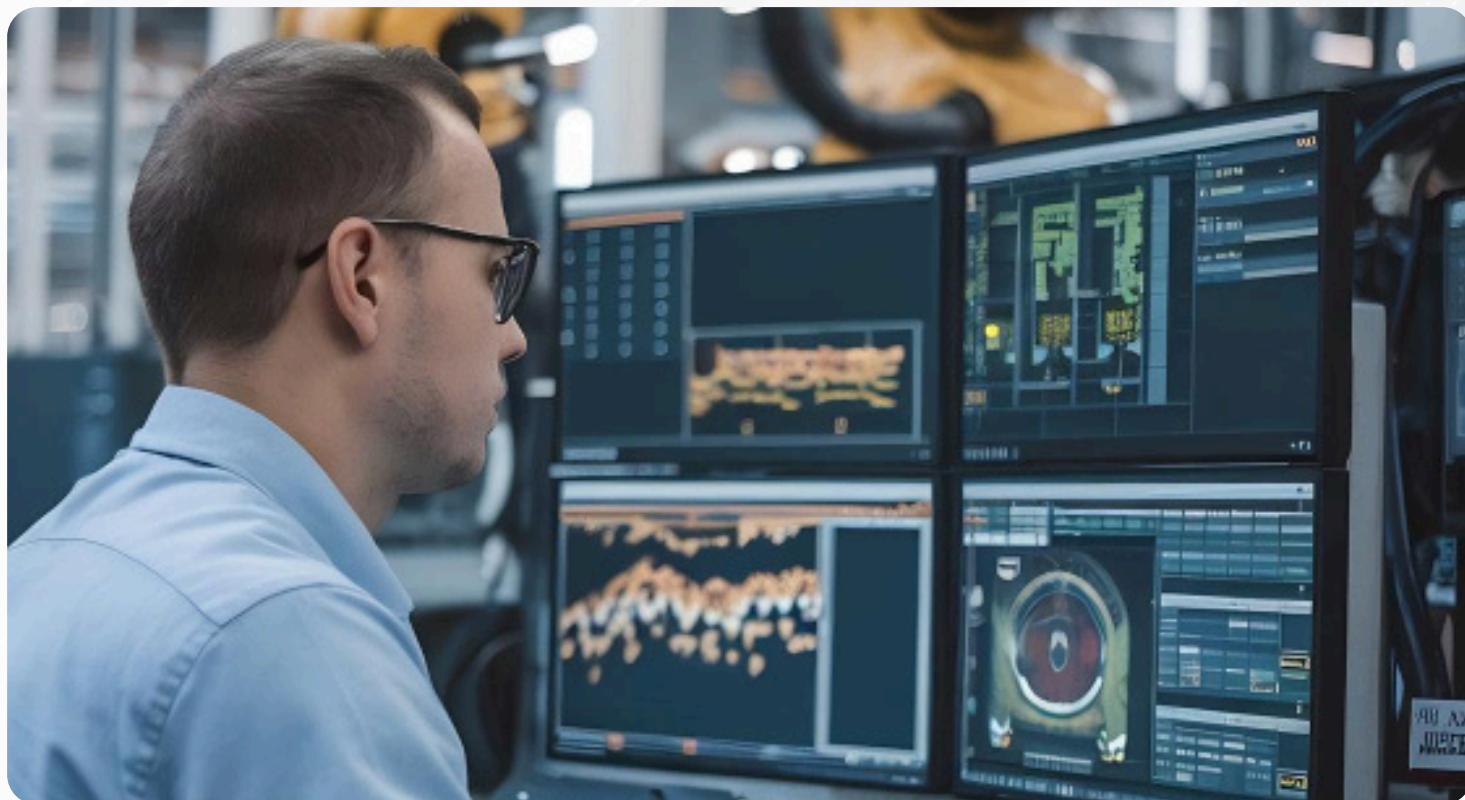
Enterprise AI + FDE Delivery (14 lessons)



- 11.1 Understanding the Forward Deployed Engineer (FDE) delivery model
- 11.2 Converting business problems into AI-driven technical solutions
- 11.3 Requirements gathering and stakeholder discovery techniques
- 11.4 Conducting effective client workshops and requirement analysis sessions
- 11.5 Problem Translation Framework for Enterprise AI projects
- 11.6 Designing AI solutions based on business objectives
- 11.7 Building rapid AI prototypes and proof-of-concepts (POCs)
- 11.8 Technical storytelling and solution presentation techniques
- 11.9 Enterprise AI project planning and delivery best practices
- 11.10 Communicating AI solutions to technical and non-technical stakeholders
- 11.11 Creating technical briefs, solution proposals, and implementation roadmaps
- 11.12 Portfolio building, case studies, GitHub projects, and LinkedIn branding
- 11.13 Real-world client engagement and consulting scenarios
- 11.14 End-to-end Enterprise AI solution delivery framework

MODULE 12

LLMOps & Production Operations (10 lessons)



- 12.1 LLMOps vs MLOps | Operating LLM systems in production — what's different and why it matters
- 12.2 Model Versioning & Registry | Track model versions; promote, roll back, and audit models (not just code)
- 12.3 Prompt & Config Management | Version prompts, feature-flag AI behavior, ship changes safely
- 12.4 Guardrails & Safety in Production | Input/output filtering, PII redaction, content moderation at the edge
- 12.5 Production Evaluation | Online eval, A/B testing live traffic, human-in-the-loop feedback loops
- 12.6 Cost Management & FinOps for AI | Token/GPU cost monitoring, budgets, rate limiting, quotas
- 12.7 Scaling & Resilience | Request queuing, caching layers, fallback models when the primary is down
- 12.8 Environment & Release Strategy | Dev / staging / prod separation, secrets rotation, disaster recovery
- 12.9 Monitoring, Drift & Incident Response | Quality drift, alerting, on-call runbooks for AI systems
- 12.10 Lab: Operate a Live AI System | Add versioning, guardrails, cost tracking & monitoring to a deployed service

Bangalore



Bhubaneswar



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